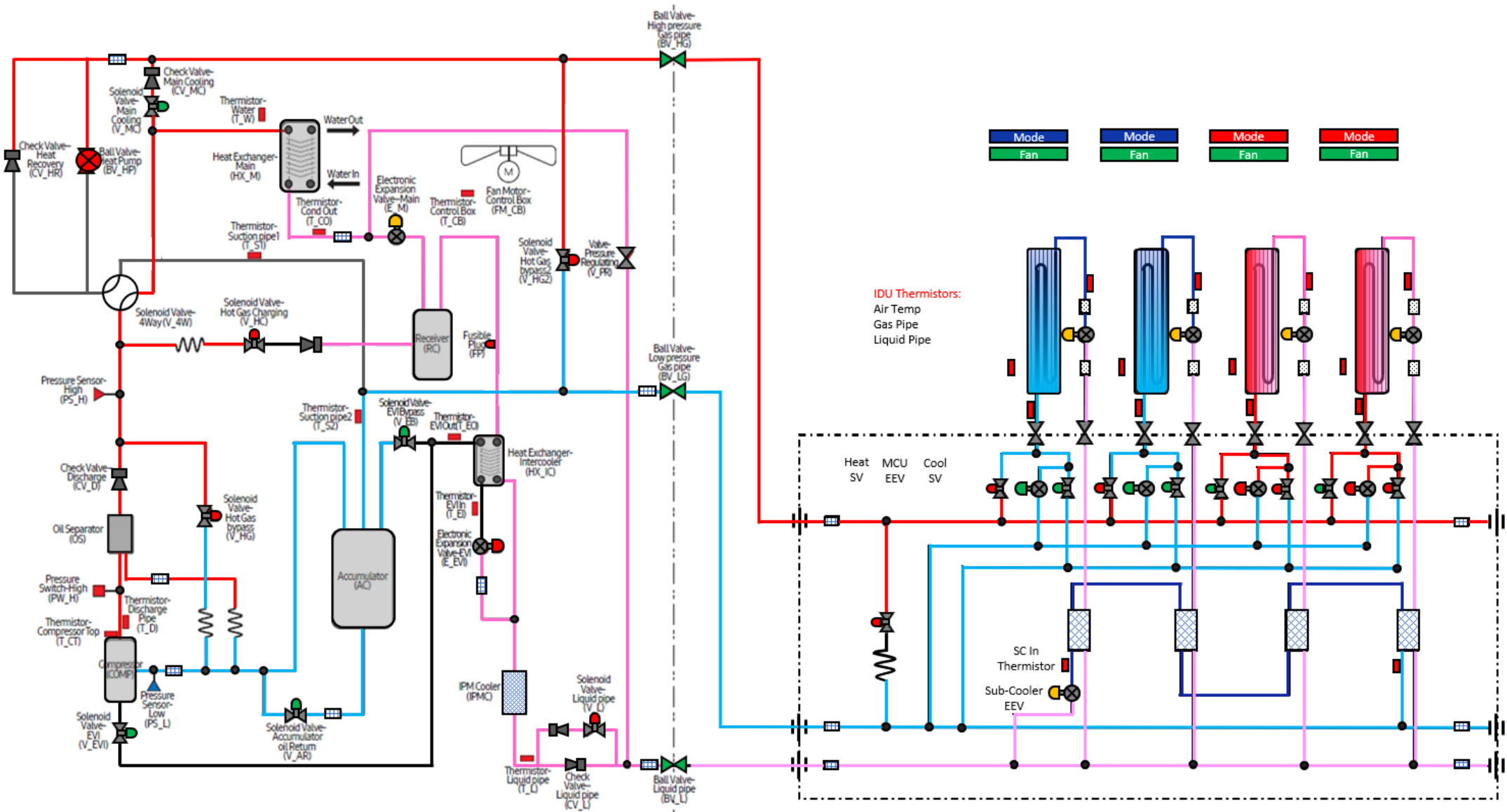


DVM S 3 Phase Water Heat Recovery
Cooling Main Mode



Comments:

Outdoor Unit		
Component	Target	Reading
Water Out Temp.		
Compressor Hz.		
High pressure		
H.P Sat. Temp		
Discharge Temp.	150 ⁰ -212 ⁰	
IPM Temp,	Below 194 ⁰	
Low pressure	101.5 psig	
L.P. Sat. Temp	32 ⁰	
Suction 1 Temp.		
Suction 2 Temp		
S.H.	0 ⁰ or 12.6 ⁰	
Min EEV Position		
Cond. Out Temp.	T_W+ (5 ⁰ – 36 ⁰)	
Liquid Line Temp.		
Sub-Cooling @ T_L	9 ⁰ - 36 ⁰	
Indoor Units		
Component	Target	Reading
IDU 1 EEV	250 – 1400 Steps	
IDU 1 SH	3.6 ⁰ (Typical 0 ⁰ -7 ⁰)	
IDU 1 T PI		
IDU 1 T PO		
IDU 2 EEV	250 – 1400 Steps	
IDU 2 SH	3.6 ⁰ (Typical 0 ⁰ -7 ⁰)	
IDU 2 T PI		
IDU 2 T PO		
IDU 3 EEV	500 – 1500 Steps	
IDU 3 SC	9 ⁰	
IDU 3 T PI		
IDU 3 T PO		
IDU 4 EEV	500 – 1500 Steps	
IDU 4 SC	9 ⁰	
IDU 4 T PI		
IDU 4 T PO		

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.



DVM S 3 Phase Water Heat Recovery

Cooling MainTarget Operation

Water Source HR Outdoor Unit				
Component	Abbreviation		Cooling Main Mode	
Compressor	Comp	Operation	Target low pressure control (default: 101.5 psig)	Primary function, Maintain Average Evaporator Temperature (Average temperature of Eva. In Sensors)
		Range	14~120Hz	Low pressure range 100psig - 128 psig. Evaporator temperature range 45 °F -48°F
High side pressure	HP	Operation	Head pressure is effected by the water temperature	Is there a variable flow control valve installed? This will effect head pressure.
		Range	299 psig - 469 psig (water temperature 68°F-104°F)	At 228psig, low limit control is activated at 512psig high limit control is activated
Main EEV	E_M	Operation	Cooling & Heating capacity balance control	
		Range	300 - 2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	SC = High Side Saturation Temp. - Liquid Line Temp.
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 104°F or 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_W < 100.4°F↓ and 65 Hz↑	Discharge Temperature Protection, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	On (Open)	Energized in cooling only
Main Cooling Solenoid	VMC	Operation	On (open)	Directs hot gas to MCU for zones calling for heat
Liquid Pipe Thermistor	T_L	Value	Used to calculate sub-cooling	Sub-Cooling range is 9°F-36°F
HP - HR Ball Valve (3 phase only)	BV_HP	Operation	Open	Valve must be open for heat pump applications
Mode Control Unit (MCU)				
Component	Abbreviation		Cooling Mode	Comments
Port Heating Solenoid	Heat	Operation	Off (Closed) ,On (Open)	Energized for units calling for heating
Port Cooling Solenoid	Cool	Operation	Off (Closed) ,On (Open)	Energized for units calling for cooling
Port EEV	EEV	Operation	Open 480 Steps	Open 480 step when a zone calls for cooling or during a mode change.
		Range	0 - 480 Steps	Closed 0 Steps when a zone is in heating
Sub-Cooling EEV	SC_EEV	Operation	Closed 0 Steps	Provide subcooling for cooling zones
		Range	480 Steps	Maintains 5.4 degrees of superheat between pipe in and out temps.
Sub-Cooler In Temperature		Operation		
Sub-Cooler Out Temperature		Operation	Close to Suction Header Temperature	
Indoor Units				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Saturation Temperature. Range 45-48°F	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Operation	Range	Low , Medium, High, Auto
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Operation	Range	Low , Medium, High, Auto
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	